Detection models/rules and scoring logic

Goal:

Your task is to build a Fraud Detection Engine that analyses payment transactions and flags suspicious operations based on a set of FRAML rules.

Steps:

1. Check input transactions against FRAML rules

Every team **must** implement these **5 mandatory** FRAML rules:

Rule id	Name	Severity	Weight
R1	CoP Name Mismatch – Hard Fail	STRONG(2)	25
R7	High frequency of transfers	STRONG(2)	10
R18	Round amounts anomaly	STRONG(2)	3
R22	Absolute High-Value Threshold	STRONG(2)	10
R1	CoP Name Mismatch – Hard Fail	MILD(1)	5

These 5 rules are mandatory. Implement the other 9 **only** after these 5 are in place.

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2. Rule Evaluation

Each transaction is evaluated by a set of rules. Each rule returns:

- **severity = 0** — rule not triggered
- **severity = 1** (MILD) — suspicion, low impact
- **severity = 2** (STRONG) — strong suspicion, high impact

Examples:

CoP hard fail (<80%) → severity 2

CoP soft mismatch (80–89%) → severity 1

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3. Risk Score Formula

```
raw_score = Σ(rule_severity × rule_weight)  //for each rule
risk_score = min(100, max(0, raw_score))    //normalization to 0–100.
```

Using severity and weights:

- Weight tells you **how important** a rule is.
- Severity tells you **how badly** the threshold was violated.

Example:

Rule: R6 High amount spike

- Weight = 8 (fixed importance)

- But the transaction may trigger it differently:
  - *slightly above threshold* → severity **1**
  - *extremely above threshold* → severity **2**

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4. Risk Categorization

Category	Example risk_score Range	Interpretation
LOW	0–29	No significant triggers — transaction OK
MEDIUM	30–69	1+ mild trigger or combinations — requires review
HIGH	70–100	Strong triggers / multiple rules — block or escalate

NOTE:

- **is_fraud_transaction** is your final decision.
- **Heuristic**: risk_score ≥ 30 = fraud (you may choose a different threshold).

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5. Generate Outputs

- **Results file** – includes all transactions
- **Alert dashboard**

Tips:

- Start with the **5 mandatory rules**
- R22, R24, R1 do not require history — the easiest to implement
- Invest time in the "history per customer" infrastructure - it will allow you to quickly add velocity/anomaly rules
- All history-based rules (R6, R7, R8, R10, R12, R13, R17, R18) filter by customer_id; only **R21** filters by customer_account (IBAN)
- R21: balance_before = balance_after + amount (when no account history is available)
- R12: use population std (divided by N, not N-1)

Example of the scoring model:

Input Transaction:

- Amount: €1,900 vs 30-day avg €400
- CoP similarity: 78%
- New beneficiary: Yes
- Destination: TR (new country)
- Fast repeat transfers: 2 in 10 min
- Balance drop: €5,000 → €3,200 (-36%)

Step 1. Apply Rule Severities (0 = none, 1 = mild, 2 = strong)

Rule	Severity	Reason
CoP Hard Fail	2	<80% similarity
High Amount Spike	1	€1,900 > 3× average
New Beneficiary + Mismatch	1	New + <90% similarity
Cross Border Anomaly	1	New corridor + >€500
Rapid Emptying	0	Drop <70%

TOTAL rule activation severity: [2, 1, 1, 1, 0]

Step 2 . Weighted Risk Score

Using simplified weights (from your scoring table):

Rule	Severity	Weight	Contribution
CoP Hard Fail	2	25	50
High Amount Spike	1	8	8
New Beneficiary + mismatch	1	10	10
Cross Border anomaly	1	8	8
Rapid emptying	0	20	0

Total = 76 (normalization not needed as < 100)

Step 3. Risk Category

Based on thresholds: 92 → HIGH RISK